

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	27 June 2026
Project Name	Opticrop:Smart Agricultural ProductionOptimization Engine
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

OptiCrop: System Architecture Diagram

Reference: [Crop Recommendation Dataset — Kaggle \(N, P, K, temperature, humidity, pH, rainfall\)](#)

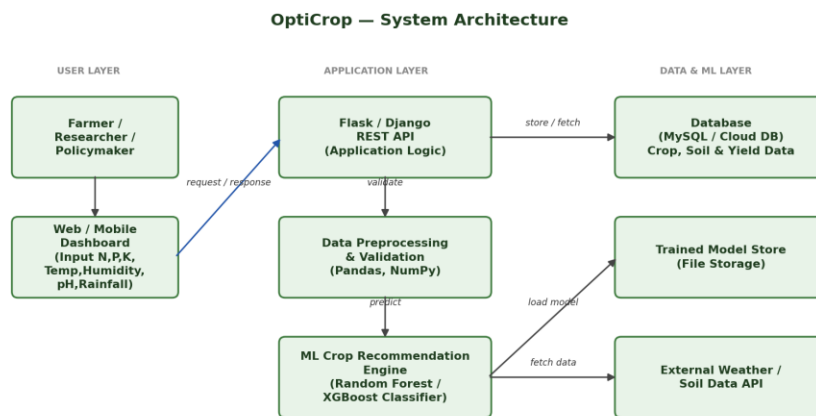


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web/mobile dashboard where farmers, researchers, and policymakers enter soil & climate parameters (N, P, K, temperature, humidity, pH, rainfall) and view crop recommendations.	HTML, CSS, JavaScript, React.js / Flask-Jinja templates
2.	Application Logic-1	Validates and preprocesses user-input soil and environmental data before passing it to the recommendation engine.	Python (Pandas, NumPy)
3.	Application Logic-2	Core ML-based crop recommendation engine that predicts the most suitable crop for the given soil & climate conditions.	Python, Scikit-learn (Random Forest / Decision Tree / XGBoost Classifier)
4.	Application Logic-3	Crop suitability & environmental assessment module that scores compatibility between input conditions and candidate crops.	Python (Flask REST API)

5.	Database	Stores historical soil, climate, crop and yield records used for training and analytics.	MySQL / PostgreSQL
6.	Cloud Database	Centralized, multi-user data store enabling researchers and policymakers to access aggregated crop-environment data.	IBM Cloudant / Firebase Firestore / MongoDB Atlas
7.	File Storage	Stores trained ML model files, datasets, and generated reports.	Local Filesystem / IBM Cloud Object Storage / AWS S3
8.	External API-1	Fetches real-time weather and rainfall data to supplement farmer-provided inputs.	OpenWeatherMap API / IBM Weather API
9.	External API-2	Integrates government soil-health and agricultural advisory data for richer recommendations.	Soil Health Card API / data.gov.in Agriculture API
10.	Machine Learning Model	Classification model trained on N-P-K, temperature, humidity, pH and rainfall data to recommend the optimal crop.	Random Forest Classifier / XGBoost (trained on crop recommendation dataset)
11.	Infrastructure (Server / Cloud)	Application deployment on local system or cloud. Local Server Configuration: Flask development server on localhost Cloud Server Configuration: Containerized deployment with auto-scaling	Local, Docker, Kubernetes, IBM Cloud Foundry / AWS EC2 / Heroku

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks used to build the web app, ML pipeline and UI.	Flask / Django, Scikit-learn, Pandas, NumPy, React.js
2.	Security Implementations	Security and access controls implemented to protect user data and the API.	HTTPS/SSL, JWT-based authentication, password hashing (bcrypt), input validation, role-based access (farmer/researcher/admin)
3.	Scalable Architecture	Architecture is designed as a 3-tier system (presentation, application/ML logic, data) so the ML inference service can scale independently.	Microservices via Docker containers, orchestrated with Kubernetes
4.	Availability	Ensures the recommendation service remains available under load or failure.	Load balancer, multiple app instances, database replication, cloud auto-scaling
5.	Performance	Design considerations to keep prediction latency low and handle concurrent requests.	Redis caching for frequent queries, optimized model inference, CDN for static assets, asynchronous request handling

